

Chapter 5: Multiple Regression Analysis

Nguyễn Thị Mộng Ngọc
University of Science, VNU - HCM
ngtmngoc@hcmus.edu.vn

Multiple Regression

- The objective is to build a probabilistic model that relates a dependent variable y to more than one independent or predictor variable.
- Let k represent the number of predictor variables ($k \geq 2$) and denote these predictors by $x_1, \dots, x_k$.
- For example,
 - $k = 2$: the gasoline mileage performance (y) of a vehicle depends on the vehicle weight (x_1) and the engine displacement (x_2).
 - $k = 3$: in attempting to predict the selling price of a house, we might have x_1 : size (ft^2), x_2 : age (years) and x_3 : number of rooms.

Definition

- The general additive **multiple regression model** equation is

$$Y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_k x_k + \varepsilon \quad (1)$$

where $\mathbb{E}[\varepsilon] = 0$ and $\mathbb{V}[\varepsilon] = \sigma^2$.

- In addition, for purposes of testing hypotheses and calculating CIs or PIs, it is assumed that ε is normally distributed.
- This is not a regression line any longer, but a regression surface.

Multiple Regression Analysis

- Let $x_1^*, x_2^*, \dots, x_k^*$ be particular values of $x_1, \dots, x_k$. Then (1) implies that

$$\mu_{Y|x_1^*, \dots, x_k^*} = \beta_0 + \beta_1 x_1^* + \dots + \beta_k x_k^*$$

Thus just as $\beta_0 + \beta_1 x$ describes the mean Y value as a function of x in simple linear regression, the **true** (or **population**) **regression function** $\beta_0 + \beta_1 x_1 + \dots + \beta_k x_k$ gives the expected value of Y as a function of $x_1, \dots, x_k$.

- The β_i 's are the **true** (or **population**) **regression coefficients**.

Multiple Regression Analysis

- The regression coefficient β_1 is interpreted as the expected change in Y associated with a 1-unit increase in x_1 while $x_2, \dots, x_k$ are held fixed.
- Analogous interpretations hold for $\beta_2, \dots, \beta_k$.
- Thus, these coefficients are called partial or adjusted regression coefficients.
- In contrast, the simple regression slope is called the marginal (or unadjusted) coefficient.

Models with Interaction and quadratic predictors

For the case of two independent variables, x_1 and x_2 , consider the following four derived models.

- 1. The first-order model:

$$Y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \varepsilon$$

- 2. The second-order no-interaction model:

$$Y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1^2 + \beta_4 x_2^2 + \varepsilon$$

- 3. The model with first-order predictors and interaction:

$$Y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1 x_2 + \varepsilon$$

- 4. The complete second-order or full quadratic model:

$$Y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1^2 + \beta_4 x_2^2 + \beta_5 x_1 x_2 + \varepsilon$$

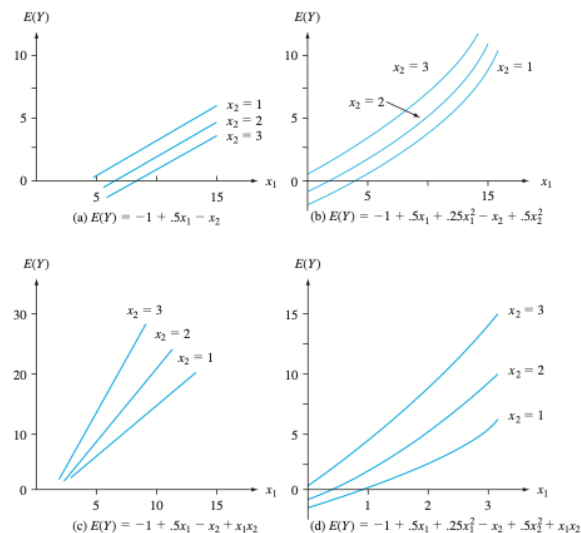


Figure. Contours of four different regression functions

Remarks

- 1. The first-order model, for a fixed value of either variable, the expected value of Y is a linear function of the other variable and that the expected change in Y associated with a unit increase in x_1 (x_2) is β_1 (β_2) independent of the level of x_2 (x_1). In Figure. (a), we obtain as contours of the regression function a collection of parallel lines, a plane in three-dimensional space.
- 2. The second-order no-interaction model, if x_2 is fixed, the expected change in Y for a 1-unit increase in x_1 is $\beta_1 + \beta_3 + 2\beta_3 x_1$. In Figure. (b), the regression surface is no longer a plane in three-dimensional space but is instead a curved surface.

Remarks

- 3. The first-order interaction model, the expected change in Y when x_1 is increased by 1 is $\beta_1 + \beta_3 x_2$. The contours of the regression function are nonparallel straight lines. In Figure. (c), this expected change depends on the value of x_2 , so each contour line must have a different slope.
- 4. The complete second-order model, the expected change in Y when x_2 is held fixed while x_1 is increased by 1 unit is $\beta_1 + \beta_3 + 2\beta_3 x_1 + \beta_5 x_2$. In Figure. (d), the contours of the regression function are both curved and not parallel to one another.

Models with predictors for categorical variables

- Thus far we have explicitly considered the inclusion of only quantitative (numerical) predictor variables in a multiple regression model.
- Using simple numerical coding, qualitative (categorical) variables can also be incorporated into a model.
- First, we should concentrate on the scenario where a variable only has two possible categories, such as male or female, U.S. or foreign-made, and so on.
- We associate a **dummy** or **indicator variable** x whose possible values 0 and 1 indicate which category is relevant for any particular observation.

Example: Travel time

- Data: y : travel time between locations in a certain city and the independent variable x_2 : distance between locations. Two types of vehicles, passenger cars and trucks, were used in the study. Let

$$x_1 = \begin{cases} 1 & \text{if the vehicle is a truck} \\ 0 & \text{if the vehicle is a passenger car} \end{cases}$$

- One possible multiple regression model is

$$Y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \varepsilon$$

- The mean value of travel time depends on whether a vehicle is a car or a truck:
 - mean time = $\beta_0 + \beta_2 x_2$ when $x_1 = 0$ (cars)
 - mean time = $\beta_0 + \beta_1 + \beta_2 x_2$ when $x_1 = 1$ (trucks)
- The coefficient β_1 is the difference in mean times between trucks and cars with distance held fixed. If $\beta_1 > 0$, on average it will take trucks longer to traverse any particular distance than it will for cars.

Example: Travel time

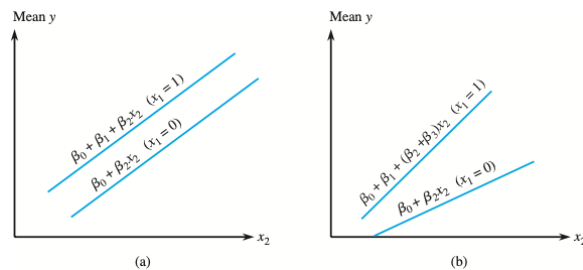
- A second possibility is a model with an interaction predictor:

$$Y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1 x_2 + \varepsilon$$

- Now the mean times for the two types of vehicles are
 - mean time = $\beta_0 + \beta_2 x_2$ when $x_1 = 0$
 - mean time = $\beta_0 + \beta_1 + (\beta_2 + \beta_3) x_2$ when $x_1 = 1$ (trucks)
- Does it make sense to have different intercepts for cars and truck? (Think about what intercept actually means).
- So, what would be a third model you could consider?

Example: Travel time

For each model, the graph of the mean time versus distance is a straight line for either type of vehicle, as illustrated in Figure 13.14.



Regression functions for models with one dummy variable (x_1) and one quantitative variable x_2 : (a) no interaction; (b) interaction
Figure 13.14

Example: Travel time

Remarks:

- The two lines are parallel for the first (no-interaction) model, but in general they will have different slopes when the second model is correct.
- For this latter model, the change in mean travel time associated with a 1-mile increase in distance depends on which type of vehicle is involved - the two variables "vehicle type" and "travel time" interact.
- Indeed, data collected by the authors of the cited article suggested the presence of interaction.

- Suppose that a multiple regression model contains two predictor variables, x_1 and x_2 . Then the data set will consist of n triples $(x_{11}, x_{21}, y_1), \dots, (x_{1n}, x_{2n}, y_n)$.

- More generally, with k predictors, the data consists of $n(k+1)$ -tuples

$$(x_{11}, x_{21}, \dots, x_{k1}, y_1), \dots, (x_{1n}, x_{2n}, \dots, x_{kn}, y_n).$$

where x_{ij} is the value of the i th predictor x_i associated with the observed value y_j .

- The observations are assumed to have been obtained independently of one another according to the model (1).
- Estimates $\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_k$ come from the minimization of the sum of square

$$f = \sum_j^n [y_j - \hat{y}_j]^2 = \sum_j^n [y_j - (\hat{\beta}_0 + \hat{\beta}_1 x_{1j} + \dots + \hat{\beta}_k x_{kj})]^2$$

Estimating Parameters

If we write things in matrix form:

$$\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}, \mathbf{X} = \begin{bmatrix} 1 & x_{11} & x_{12} & \cdots & x_{1k} \\ 1 & x_{21} & x_{22} & \cdots & x_{2k} \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & x_{n1} & x_{n2} & \cdots & x_{nk} \end{bmatrix}, \boldsymbol{\beta} = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_k \end{bmatrix}, \boldsymbol{\epsilon} = \begin{bmatrix} \epsilon_1 \\ \epsilon_2 \\ \vdots \\ \epsilon_n \end{bmatrix}$$

Then we have (1) written as

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon},$$

and the Least Squares estimate $\hat{\boldsymbol{\beta}}$ of $\boldsymbol{\beta}$ has a closed form expression

$$\hat{\boldsymbol{\beta}} = \begin{bmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \\ \vdots \\ \hat{\beta}_k \end{bmatrix} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}.$$

Example 8

- Data: $n = 30$ with $x_1 = \text{force (gm)}$, $x_2 = \text{power (mW)}$, $x_3 = \text{tempertaure (oC)}$, and $x_4 = \text{time (msec)}$ on $y = \text{ball bond shear strength (gm)}$.
- The data can be read and fitted as following.

```
> dim(xmp13.12)
[1] 30 6
> head(xmp13.12,2)
  Observation Force Power Temperature Time Strength
1           1    30   60          175    15    26.2
2           2    40   60          175    15    26.3

> fit=lm(y~x1+x2+x3+x4)
> coef(fit)
(Intercept)          x1          x2          x3          x4
-37.4766667    0.2116667    0.4983333    0.1296667    0.2583333
```

Output in R:

```
> summary(fit)

Call:
lm(formula = y ~ x1 + x2 + x3 + x4)

Residuals:
    Min       1Q   Median       3Q      Max
-11.0900  -1.7608  -0.3067   2.4392   7.5933

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -37.47667   13.09964  -2.861  0.00841 **
x1           0.21167    0.21057   1.005  0.32444
x2           0.49833    0.07019   7.100 1.93e-07 ***
x3           0.12967    0.04211   3.079 0.00499 **
x4           0.25833    0.21057   1.227 0.23132
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 5.158 on 25 degrees of freedom
Multiple R-squared:  0.714, Adjusted R-squared:  0.6682
F-statistic: 15.6 on 4 and 25 DF, p-value: 1.592e-06
```

Example 8 (continued)

This output gives the values of the least squares estimates

$$\begin{pmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \beta_3 \\ \beta_4 \end{pmatrix} = \begin{pmatrix} -37.47667 \\ 0.21167 \\ 0.49833 \\ 0.12967 \\ 0.25833 \end{pmatrix}$$

Therefore, the fitted least squares regression model is

$$\hat{y} = -37.48 + 0.2117 * x_1 + 0.4983 * x_2 + 0.1297 x_3 + 0.2583 * x_4$$

or, in other words,

$$\hat{strength} = -37.48 + 0.2117 * force + 0.4983 * power + 0.1297 * temperture + 0.2583 * time$$

- A point prediction of strength resulting from a force of 35 gm, power of 75 mW, temperature of 200° degrees, and time of 20 msec is

$$\hat{y} = -37.48 + 0.2117 * 35 + 0.4983 * 75 + 0.1297 * 200 + 0.2583 * 20 = 38.41$$

- This is also a point estimate of the mean value of strength for the specified values of force, power, temperature, and time.

- Output in R:

```
> fit$fitted.values
 1      2      3      4      5      6      7      8      9
25.34000 27.45667 40.29000 42.40667 31.82333 33.94000 46.77333 48.89000 27.92333 30.04000
13     14     15     16     17     18     19     20     21
34.40667 36.52333 49.35667 51.47333 36.29000 40.52333 23.45667 53.35667 31.92333 44.89000
25     26     27     28     29     30
38.40667 38.40667 38.40667 38.40667 38.40667 38.40667
> fit$fitted.values[30]
[1] 38.40667
> xmp13.11[30,]
  Observation Force Power Temperature Time Strength
30           30    35   75          200    20    40.3
```

Estimating the error variance

- Define the error or residual sum of squares

$$SSE = \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \sum_{i=1}^n e_i^2$$

- In the multiple linear regression model

$$Y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_k x_k + \varepsilon \quad \text{where } \varepsilon \sim \mathcal{N}(0, \sigma^2).$$

$$\hat{\sigma}^2 = s^2 = \frac{SSE}{n - (k + 1)} = MSE$$

- The residual standard error is

$$\hat{\sigma} = s = \sqrt{MSE} = \sqrt{\frac{SSE}{n - (k + 1)}}$$

- IDENTITY : Algebraically, it can be shown that

$$\sum_{j=1}^n (y_i - \bar{y})^2 = \sum_{j=1}^n (\hat{y}_i - \bar{y})^2 + \sum_{j=1}^n (y_i - \hat{y}_i)^2$$

$$SST = SSR + SSE$$

- $SSE = \sum_{j=1}^n (y_i - \hat{y}_i)^2$ is a measure of how much variation in the observed y values is not explained by the model relationship.
- $SST = \sum_{j=1}^n (y_i - \bar{y})^2$, Total sum of squares, is a measure of total variation in the observed y values.
- $SSR = \sum_{j=1}^n (\hat{y}_i - \bar{y})^2$, Regression sum of squares, is a measure of explained variation.

ANOVA table

We can combine all of this information to produce an analysis of variance (ANOVA) table. Such tables are standard in regression analysis.

Source	Sum of Squares	Degrees of Freedom	Mean Square	F
Regression	SSR	k	MSR	$\frac{MSR}{MSE}$
Error	SSE	n-(k+1)	MSE	
Total	SST	n-1		

The coefficient of multiple determination R^2

- The coefficient of multiple determination R^2 is

$$R^2 = 1 - \frac{SSE}{SST} = \frac{SSR}{SST}$$

- Clearly,

$$0 \leq R^2 \leq 1$$

The larger the R^2 , the better the regression model explains the variability in the data.

The adjusted R^2

One draw back of R^2 is that, it can never decrease when a new x is added. Thus, it can be difficult to judge whether the increase is telling us anything useful about the new x . Instead, many regression users prefer to use an **adjusted R^2** statistic:

$$R_a^2 = 1 - \frac{SSE/(n - (k + 1))}{SST/(n - 1)}$$

- R_a^2 is smaller than R^2 .
- **Adjusted R^2 can even be negative**, whereas R^2 itself must be between 0 and 1.
- A value of R_a^2 that is substantially smaller than R^2 itself is a warning that the model may contain too many predictors.

Example 9

- Data: $n = 9$, $x_1 = \%$ limestone powder, $x_2 = \text{water-cement ratio}$ and $y = \text{compressive strength}$.
- The data can be read as following

```
> dim(xmp13.13)
[1] 9 5
> head(xmp13.13,2)
  x1  x2  x1x2  X28 Absorbability
1 21 0.65 13.65 33.55          8.42
2 21 0.55 11.55 47.55          6.26
```

- **Fitting the first-order model results in**

$$y = 84.8167 + 0.1643 * x_1 - 79.6667 * x_2, R^2 = 0.7406, R_a^2 = 0.6541$$

```
> fit = lm(y~x1+x2 )
> summary(fit)
```

Call:

```
lm(formula = y ~ x1 + x2)
```

Residuals:

```
      Min       1Q   Median       3Q      Max
-6.2500 -0.7167  0.7167  2.0833  3.1000
```

Coefficients:

```
      Estimate Std. Error t value Pr(>|t|)
(Intercept)  84.8167    12.2415   6.929 0.000448 ***
x1           0.1643     0.1431   1.148 0.294673
x2          -79.6667    20.0349  -3.976 0.007313 **
---

```

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.47 on 6 degrees of freedom
Multiple R-squared: 0.7406, Adjusted R-squared: 0.6541
F-statistic: 8.565 on 2 and 6 DF, p-value: 0.01746

Output in R continued

- **Fitting the first-order model results in**

$$y = 84.8167 + 0.1643 * x_1 - 79.6667 * x_2, R^2 = 0.7406, R_a^2 = 0.6541 \text{ and } SSE = 72.252(df = 6).$$

```
> fit = lm(y~x1+x2 )
> anova(fit)
```

Analysis of Variance Table

Response: y

```
      Df Sum Sq Mean Sq F value    Pr(>F)
x1      1  15.870   15.870   1.3179 0.294673
x2      1 190.403  190.403  15.8117 0.007313 **
Residuals  6  72.252   12.042
---

```

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Fitting the model with first-order predictors and interaction:

$$y = 6.217 + 5.779x_1 + 51.333x_2 - 9.357x_1x_2, R^2 = 0.8946, R_a^2 = 0.8314$$

```
> summary(fit_inter)
```

```
Call:
```

```
lm(formula = y ~ x1 + x2 + x1x2)
```

```
Residuals:
```

```
1      2      3      4      5      6      7      8      9
0.3417 -0.1750 -2.4583 -2.9750 -0.7167  2.0833  0.2000  0.7167  2.9833
```

```
Coefficients:
```

```
Estimate Std. Error t value Pr(>|t|)
(Intercept)  6.217      30.304   0.205   0.8455
x1           5.779       2.079   2.779   0.0389 *
x2          51.333      50.434   1.018   0.3555
x1x2        -9.357       3.461  -2.704   0.0426 *
```

```
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 2.423 on 5 degrees of freedom
Multiple R-squared:  0.8946, Adjusted R-squared:  0.8314
F-statistic: 14.15 on 3 and 5 DF, p-value: 0.007061
```

Fitting the model with first-order predictors and interaction:

$$\hat{y} = 6.217 + 5.779x_1 + 51.333x_2 - 9.357x_1x_2, R^2 = 0.8946, R_a^2 = 0.8314, SSE = 29.349(df = 5)$$

```
> fit_inter=lm(y~x1+x2+x1x2 )
```

```
> anova(fit_inter)
```

```
Analysis of Variance Table
```

```
Response: y
```

```
Df Sum Sq Mean Sq F value Pr(>F)
x1    1  15.870    15.870    2.7037 0.161039
x2    1 190.403   190.403   32.4376 0.002328 **
x1x2   1  42.903    42.903    7.3090 0.042605 *
Residuals  5  29.349     5.870
```

```
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Prediction for compressive strength when

% limestone = 14 and water-cement ratio = 0.60 (9th observation) is

$$\hat{y} = 6.217 + 5.779 \cdot 14 + 51.333 \cdot 0.6 - 9.357 \cdot 14 \cdot 0.6 = 39.324.$$

```
> fit_inter$fitted.values[9]
```

```
39.31667
```

F statistics

The F statistic in the ANOVA table is used to test

- Hypotheses:

$$H_0: \beta_1 = \beta_2 = \dots = \beta_k = 0$$

$$H_1: \text{at least one } \beta_i \neq 0 \quad (i = 1, \dots, k)$$

- Test statistic value

$$f = \frac{R^2/k}{(1-R^2)/(n-(k+1))} = \frac{SSR/k}{(SSE)/(n-(k+1))} = \frac{MSR}{MSE}$$

When H_0 is true, the test statistic F has an $\mathcal{F}$ distribution with k numerator df and $n - (k + 1)$ denominator df.

- The test is upper-tailed, so the p-value is the area under the $F_{k, n-(k+1)}$ curve to the right of f.

Back to Example 8, a model with $k = 4$ predictors was fit, so the relevant hypotheses are

$$H_0: \beta_1 = \beta_2 = \beta_3 = \beta_4 = 0$$

$$H_1: \text{at least one } \beta_i \neq 0 \quad (i = 1, \dots, 4)$$

Output in R:

```
> fit=lm(y~x1+x2+x3+x4)
```

```
> simpleAnova(fit)
```

```
Analysis of Variance Table
```

```
Response: y
```

```
Df Sum Sq Mean Sq F value Pr(>F)
Predictors  4 1660.14   415.03    15.6 1.592e-06 ***
Residuals  25  665.12    26.60
```

```
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

This value appears in the F value column of the ANOVA table: $f = 15.6$ and $p_{value} = 1.592e - 06 < 0.001$. This is a highly significant result. The null hypothesis should be rejected at any reasonable significance level.

- Confidence Intervals for β_i A $100(1 - \alpha)\%$ confidence intervals (CI) for β_i , the coefficient of x_i with $i = 0, 1, \dots, k$ in the regression function, is

$$\hat{\beta}_i \pm t_{\alpha/2, n-(k+1)} * s_{\hat{\beta}_i}$$

- A test for $H_0 : \beta_i = 0$ vs $H_0 : \beta_i \neq 0$ uses the t statistic value $t = \frac{\hat{\beta}_i}{s_{\hat{\beta}_i}}$ based on $n - (k + 1)$ df.

You can be performed by examining the p_{value} output provided in R (Remark: H_0 will be rejected if $p_{value} \leq \alpha$).

- If $H_0 : \beta_i = 0$ is not rejected, then x_i is not important in describing Y in the presence of the other independent variables.
- If $H_0 : \beta_i = 0$ is rejected, this means that x_i is important in describing Y even after including the effects of the other independent variables.

- Hypotheses:

$$H_0 : \beta_1 = \beta_2 = \dots = \beta_k = 0$$

$$H_1 : \text{at least one } \beta_j \neq 0 \quad (j = 1, \dots, k)$$

- Test statistic value

$$f = \frac{R^2/k}{(1 - R^2)/(n - (k + 1))} = \frac{SSR/k}{(SSE)/(n - (k + 1))} = \frac{MSR}{MSE}$$

When H_0 is true, the test statistic F has an $\mathcal{F}$ distribution with k numerator df and $n - (k + 1)$ denominator df.

- The test is upper-tailed, so the p -value is the area under the $F_{k, n-(k+1)}$ curve to the right of f.

- A $100(1 - \alpha)\%$ CI for $\mu_{Y|x_1^*, \dots, x_k^*}$, the expected value of Y when $x_1, \dots, x_k = x_1^*, \dots, x_k^*$, is

$$\hat{\mu}_{Y|x_1^*, \dots, x_k^*} \pm t_{\alpha/2, n-(k+1)} s_{\hat{\mu}_{Y|x_1^*, \dots, x_k^*}} = \hat{y} \pm t_{\alpha/2, n-(k+1)} s_{\hat{y}}$$

where $\hat{Y}$ is the statistic $\hat{\beta}_0 + \hat{\beta}_1 x_1^*, \dots, \hat{\beta}_k x_k^*$ and $\hat{y}$ is the calculated value of $\hat{Y}$.

- A $100(1 - \alpha)\%$ Prediction Interval (PI) for a future y value

$$\hat{\mu}_{Y|x_1^*, \dots, x_k^*} \pm t_{\alpha/2, n-(k+1)} \sqrt{s^2 + s_{\hat{\mu}_{Y|x_1^*, \dots, x_k^*}}^2} = \hat{y} \pm t_{\alpha/2, n-(k+1)} \sqrt{s^2 + s_{\hat{y}}^2}$$

Simultaneous intervals for which the simultaneous confidence or prediction level is controlled can be obtained by applying the Bonferroni technique.

- Data: on microhardness (GPa) $n = 18$,

- Model: $Y = \beta_0 + \beta_1 * N + \beta_2 * F + \varepsilon$

- Output from R

```
> summary(fit)

Call:
lm(formula = microhardness ~ N + F)

Residuals:
    Min       1Q   Median       3Q      Max
-0.14798 -0.06838  0.01103  0.05453  0.17458

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  6.227686   0.046155  134.930 < 2e-16 ***
N             0.061823   0.002099   29.459 1.09e-14 ***
F            -0.038721   0.011221   -3.451  0.00357 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.1001 on 15 degrees of freedom
Multiple R-squared:  0.9838, Adjusted R-squared:  0.9817
F-statistic: 456.3 on 2 and 15 DF, p-value: 3.678e-14
```

- The very high R^2 ($R^2 = 0.9838$) indicates that almost all of the observed variation in micro-hardness can be attributed to the model relationship and the fact that both nitrogen % and fluorine % are varying.

- We test $H_0 : \beta_1 = 0$ vs $H_0 : \beta_1 \neq 0$
- The t statistic value
 $t = \frac{\beta_i}{s_{\beta_i}} = 0.061823/0.002099 = 29.45355 < 2.131 = t_{0.025,15}$. We have $p_{value} = 1.09e - 14$ H_0 would be rejected at any reasonable significance level.
- Output from R

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	6.227686	0.046155	134.930	< 2e-16 ***
N	0.061823	0.002099	29.459	1.09e-14 ***
F	-0.038721	0.011221	-3.451	0.00357 **

 Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
> fit=lm(microhardness ~N+F)
> anova(fit)
Analysis of Variance Table
```

```
Response: microhardness
      Df Sum Sq Mean Sq F value    Pr(>F)
N       1  9.0157   9.0157 900.646 8.271e-15 ***
F       1  0.1192   0.1192  11.907 0.003567 **
Residuals 15 0.1502   0.0100
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
> simpleAnova(fit)
Analysis of Variance Table
```

```
Response: microhardness
      Df Sum Sq Mean Sq F value    Pr(>F)
Predictors  2  9.1348   4.5674  456.28 3.678e-14 ***
Residuals  15 0.1502   0.0100
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

- We test

$$H_0 : \beta_1 = \beta_2 = \dots = \beta_k = 0$$

$$H_1 : \text{at least one } \beta_j \neq 0 \quad (j = 1, \dots, k)$$

- The f statistic value $f = \frac{MSR}{MSE} = 456.28$ and $p_{value} = 3.678e - 14$, H_0 would be rejected at any reasonable significance level.
- Output from R

```
> simpleAnova(fit)
Analysis of Variance Table
```

```
Response: microhardness
      Df Sum Sq Mean Sq F value    Pr(>F)
Predictors  2  9.1348   4.5674  456.28 3.678e-14 ***
Residuals  15 0.1502   0.0100
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

- When $N = 20$ and $F = 1$ then

$$\hat{y} = 6.227686 + 0.061823 * 20 - 0.038721 * 1 = 7.4254$$

- Output from R

```
> data[16,]
      N      F microhardness
    20.0     1.0           7.6
> fit$fitted.values
      1      2      3      4      5      6      7      8      9     10
6.227686 7.464147 6.188965 7.425425 8.661886 6.034079 6.343194 6.652309 6.961424 7.270540
    11     12     13     14     15     16     17     18
6.227686 6.188965 6.111522 6.034079 7.464147 7.425425 7.347982 7.270540
> fit$fitted.values[16,]
Error in fit$fitted.values[16, ] : incorrect number of dimensions
> fit$fitted.values[16]
      16
7.425425
```

- A 95% CI for β_1 :

$$\hat{\beta}_1 \pm t_{0.025, 15} s_{\hat{\beta}_1} = 0.061823 \pm 2.131 * 0.002099 \approx (0.0574, 0.0663)$$

- A 95% CI for β_2 :

$$\hat{\beta}_2 \pm t_{0.025, 15} s_{\hat{\beta}_2} = -0.038721 \pm 2.131 * 0.011221 \approx (-0.0626, -0.0148)$$

- Since neither 95% CI contains 0, it is no surprise that both null hypotheses are rejected at significance level 5%.

- Output from R

```
> #####confidence intervals for $beta_i$
> confint(fit, level=0.95)
              2.5 %      97.5 %
(Intercept)  6.12930918  6.32606272
N             0.05734999  0.06629609
F            -0.06263915 -0.01480373
```

- A 95% CI for true average hardness when $N = 20$ and $F = 1$:

$$\hat{y} \pm t_{\alpha/2, n-(k+1)} s_{\hat{y}} = 7.4254 \pm 2.131 * 0.0332 \approx (7.35, 7.50)$$

- A 95% PI for the hardness resulting from a single observation when $N = 20$ and $F = 1$:

$$\hat{y} \pm t_{\alpha/2, n-(k+1)} \sqrt{s^2 + s_{\hat{y}}^2} = 7.42546 \pm 2.131 \sqrt{(0.100051)^2 + (0.0332)^2} \approx (7.20, 7.65)$$

- Output from R

```
> #####Confidence interval for the mean response
> predict.lm(fit, new=data.frame(N=20, F=1), interval="confidence")
              fit      lwr      upr
1 7.425425 7.35472 7.496131
> #####Prediction Interval for New Observations
> predict(fit, new=data.frame(N=20, F=1), interval = c("prediction"))
              fit      lwr      upr
1 7.425425 7.200756 7.650095
```

Partial F-test: Let's label the predictors as $x_1, x_2, \dots, x_l, x_{l+1}, \dots, x_k$, so that it is the last $k - l$ that we are considering deleting.

- Hypotheses:

$$H_0: \beta_{l+1} = \beta_{l+2} = \dots = \beta_k = 0$$

(so the "reduced" model $Y = \beta_0 + \beta_1 x_1 + \dots + \beta_l x_l + \varepsilon$ is correct).

$$H_1: \text{at least one among } \beta_{l+1}, \beta_{l+2}, \dots, \beta_k \text{ is not } 0$$

(so in the "full" model $Y = \beta_0 + \beta_1 x_1 + \dots + \beta_k x_k + \varepsilon$).

- Test statistic value

$$f = \frac{(SSE_l - SSE_k)/(k - l)}{SSE_k/(n - (k + 1))}$$

where SSE_k : unexplained variation for the full model
and SSE_l : unexplained variation for the reduced model.

- The test is upper-tailed; the p -value is the area under the $F_{k-l, n-(k+1)}$ curve to the right of f .

Example 11

- Data:

- on microhardness (GPa) $n = 17$,
- y = SDF content (%) in soybean residue
- x_1 = extrusion temperature(°C)
- x_2 = feed moisture %)
- x_3 = screw speed (rpm)

- The complete second-order model :

$$Y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_1^2 + \beta_5 x_2^2 + \beta_6 x_3^2 + \beta_7 x_1 x_2 + \beta_8 x_1 x_3 + \beta_9 x_2 x_3 + \varepsilon$$

- The reduced model:

$$Y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \varepsilon$$

```
> summary(lmAll)

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -1.316e+02  1.041e+01 -12.642 4.48e-06 ***
x1           1.688e+00  2.764e-01   6.104 0.000489 ***
x2           7.769e-01  6.683e-02  11.624 7.87e-06 ***
x3           7.979e-01  8.484e-02   9.405 3.20e-05 ***
x1sqd       -2.700e-02  3.418e-03  -7.899 9.88e-05 ***
x2sqd       -2.756e-03  2.136e-04 -12.902 3.90e-06 ***
x3sqd       -2.037e-03  2.136e-04  -9.537 2.92e-05 ***
x1x2        -8.750e-04  8.767e-04  -0.998 0.351510
x1x3         6.000e-04  8.767e-04   0.684 0.515759
x2x3        -6.187e-04  2.192e-04  -2.823 0.025663 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.1753 on 7 degrees of freedom
Multiple R-squared:  0.9872, Adjusted R-squared:  0.9707
F-statistic: 59.93 on 9 and 7 DF,  p-value: 8.563e-06
```

- Note that the $R^2 = 0.987$, so almost all of the observed variation in y can be explained by the model relationship.
- The F ratio for model utility is 59.93 with a corresponding $p_{value} = 8.563e - 06$. So the null hypothesis that all nine β_i 's corresponding to predictors have value 0 is resoundingly rejected. There appears to be a useful relationship between the dependent variable and at least one of the predictors.

Is the inclusion of the second-order predictors justified?

- The hypotheses to be tested are

$$H_0 : \beta_4 = \beta_5 = \dots = \beta_9 = 0$$

$$H_1 : \text{at least one among } \beta_4, \beta_5, \dots, \beta_9 \text{ is not } 0$$

- The value of the F statistic

$$f = \frac{(11.4281 - .2152)/(9 - 3)}{.2152/(17 - (9 + 1))} = 60.78865 < F_{\alpha, 6, 7}$$

or $p_{value} = 1.101e - 05$, So the null hypothesis is resoundingly rejected.

- Output from R

```
> anova(lmH0, lmAll)
Analysis of Variance Table

Model 1: y ~ x1 + x2 + x3
Model 2: y ~ x1 + x2 + x3 + x1sqd + x2sqd + x3sqd + x1x2 + x1x3 + x2x3
  Res.Df  RSS Df Sum of Sq    F    Pr(>F)
1      13 11.4281
2       7  0.2152  6    11.213 60.782 1.101e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```